

BITBURG, Germany

WMO: 106100

Lat: 49.950N

Lon: 6.567E

Elev: 374

StdP: 96.91

Time Zone: 1.00 (EUC)

Period: 82-93

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.8	-7.9	-15.1	1.1	-9.8	-12.8	1.3	-6.7	13.2	7.3	11.8	5.2	4.4	60	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	28.2	18.8	26.9	18.7	24.9	17.4	20.0	26.3	19.2	25.2	18.2	23.7	3.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.5	22.3	17.0	12.7	21.6	16.0	11.9	20.9	59.5	26.6	56.1	25.0	52.9	23.6	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 10.1	8.8	7.9		-11.3	32.5	3.9	2.0	-14.1	34.0	-16.3	35.1	-18.5	36.2	-21.3		
(5)				-12.0	22.0	3.7	1.2	-14.6	22.9	-16.8	23.6	-18.9	24.2	-21.6	25.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	0.9	0.6	4.7	7.8	12.1	14.9	17.9	17.7	14.1	9.8	4.5	2.4		
	DBStd	7.11	4.61	4.45	3.65	3.62	3.66	3.37	3.29	3.27	3.05	3.41	3.50	3.66		
	HDD10.0	1276	283	264	169	86	20	2	0	0	3	47	168	236		
	HDD18.3	3491	541	497	424	317	193	112	50	51	131	266	416	494		
	CDD10.0	911	0	1	4	19	86	150	246	240	124	39	2	1		
	CDD18.3	82	0	0	0	0	1	9	37	32	3	0	0	0		
	CDH23.3	749	0	0	0	1	17	78	330	287	36	0	0	0		
	CDH26.7	169	0	0	0	0	2	8	86	71	3	0	0	0		
(14) Wind	WSAvg	3.3	4.3	3.7	3.9	3.3	3.0	2.9	2.8	2.6	2.9	3.3	3.4	3.5		
(15) Precipitation	PrecAvg	767	70	57	50	47	65	70	71	66	62	65	66	79		
	PrecMax	1009	197	149	141	123	128	171	194	144	138	183	133	217		
	PrecMin	565	8	5	7	0	13	20	22	10	17	23	4	10		
	PrecStd	109	38	34	32	26	31	37	36	34	35	34	30	42		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	13.0	18.0	21.2	25.1	27.9	31.2	31.8	27.0	21.2	13.1	12.2		
		MCWB	8.5	8.5	11.2	13.4	15.8	19.5	19.9	19.2	17.6	16.0	11.5	10.3		
	2%	DB	9.1	10.1	14.8	18.9	23.1	25.9	29.0	28.2	24.0	18.1	12.0	10.8		
		MCWB	7.8	7.5	9.1	11.7	15.5	18.0	19.2	18.7	17.5	14.6	10.3	9.0		
	5%	DB	8.0	8.1	12.1	16.2	21.2	23.2	27.0	26.8	21.8	16.2	10.9	8.9		
		MCWB	7.2	6.2	8.4	10.4	14.3	16.9	19.0	18.6	15.9	13.7	9.5	7.8		
	10%	DB	7.0	6.2	10.1	14.1	19.1	21.1	24.9	24.2	19.8	14.8	9.2	7.2		
		MCWB	6.0	4.7	7.2	9.1	13.5	16.0	17.6	17.3	15.1	12.5	8.0	6.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.3	9.6	11.7	14.3	17.3	21.2	21.5	21.0	19.5	16.6	12.2	10.7		
		MCDB	10.1	12.4	16.3	20.5	23.5	25.4	28.3	28.3	23.9	19.8	13.0	11.7		
	2%	WB	8.0	7.9	9.9	12.4	16.0	19.1	20.1	19.7	17.9	15.3	10.6	9.1		
		MCDB	8.7	9.9	13.6	17.6	21.6	23.8	27.0	26.3	22.4	17.9	11.7	10.3		
	5%	WB	7.1	6.2	8.8	10.9	14.9	17.8	19.3	18.8	16.7	14.0	9.7	7.8		
		MCDB	7.9	7.7	11.7	15.3	20.0	22.1	25.6	25.1	20.5	16.0	10.8	8.7		
	10%	WB	5.8	4.8	7.5	9.8	13.9	16.5	18.3	17.9	15.6	12.9	8.4	6.6		
		MCDB	6.6	6.1	9.8	13.5	18.6	20.6	24.0	23.4	19.0	14.7	9.2	7.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	5.6	7.1	8.7	9.2	8.5	10.0	10.0	8.6	6.8	5.0	4.1		
		MCDBR	5.6	7.8	10.8	12.1	12.6	11.9	13.4	13.4	12.4	9.1	6.2	5.4		
	5% WB	MCWBR	5.1	6.3	7.1	7.1	7.0	6.6	6.4	6.2	7.3	6.2	5.6	5.1		
		MCWBR	5.4	7.0	9.5	11.0	11.3	10.6	12.3	11.9	10.5	8.7	5.5	4.9		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.328	0.359	0.390	0.399	0.401	0.399	0.395	0.367	0.350	0.324	0.300		
		taud	2.436	2.387	2.322	2.268	2.274	2.288	2.287	2.327	2.385	2.429	2.440	2.427		
	Ebn at Noon		747	810	842	852	859	858	853	838	826	773	709	691		
		Edh at Noon	62	82	104	123	127	127	125	114	97	79	62	55		
(44) All-Sky Solar Radiation	RadAvg		0.79	1.50	2.67	4.21	4.96	5.51	5.22	4.47	3.34	1.93	0.89	0.61		
	RadStd		0.12	0.28	0.41	0.52	0.50	0.50	0.57	0.45	0.41	0.19	0.11	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page